

Training Multimodal Systems

Three-stage pipeline, pre-training objectives, data curation, and the \$300 LLaVA-1.5 recipe

10 min

Duration

B3 – Apply

Bloom's

L3 – Intermediate

Level

TH-000010 & 000011

Prerequisite

Session Overview

What you will learn and why training multimodal systems is fundamentally different from training text LLMs

What You Will Learn

- The four core challenges unique to multimodal training
- Stage 1: Unimodal pre-training (each encoder separately)
- Stage 2: Cross-modal alignment pre-training (CLIP, BEiT-3)
- Stage 3: Instruction tuning (LLaVA-Instruct, ShareGPT4V)
- Contrastive, generative, and hybrid pre-training objectives
- Data curation: why quality beats quantity (DataComp insight)
- Curriculum learning and modality balancing
- LLaVA-1.5 training recipe: exact numbers, cost, and timeline

Concept at a Glance

- Duration: 10 minutes
- Bloom's Level: B3 – Apply
- Difficulty: L3 – Intermediate
- Module: 4.3 Multimodal Fusion and Training
- Prerequisites: Fusion Strategies (TH-000010), Audio-Visual Learning (TH-000011)
- Next: Multimodal Fine-Tuning (TH-000013)

Learning Objectives

By the end of this session you will be able to:

10 min

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TH-000010 & 000011

Prerequisite

- 1 Describe the multi-stage training pipeline common to all large multimodal models
- 2 Distinguish contrastive, generative, and matching pre-training objectives and name models for each
- 3 Explain why data curation is as important as model architecture (DataComp insight)
- 4 Identify the main pre-training datasets: LAION-5B, CC12M, WebVid-10M, AudioSet
- 5 Apply the concept of curriculum learning to multimodal training
- 6 Trace the LLaVA-1.5 two-stage training recipe with exact parameters and costs

SECTION 1 — WHY MULTIMODAL TRAINING IS DIFFERENT

Four Unique Challenges

Adding modalities multiplies challenges across data, architecture, optimisation, and compute

The Four Core Challenges of Multimodal Training

Training a text LLM is already complex. Each modality adds a new data pipeline, encoder, and alignment requirement

Data Heterogeneity

Text: trillions of tokens available. Image-text pairs: billions. Video-text pairs: hundreds of millions. Audio-visual pairs: tens of millions. This 100,000× imbalance means the model's text capabilities will dominate unless explicitly balanced.

Representation Alignment

A ViT patch embedding and a language model token embedding are fundamentally incompatible vectors — different dimensionality, different distribution, different semantic space. A bridge (projector, Q-Former) must be trained to translate between them.

Training Stability

Multiple encoders, multiple loss functions, and gradient flow across modality boundaries create instability. Loss spikes in one modality can corrupt the entire training run. Gradient magnitude equalisation and careful loss weighting are essential.

Compute Scaling

A 336×336 image → 576 ViT tokens. A 1-minute video at 1 fps → 3,600 tokens. Quadratic attention complexity means visual sequences dominate memory. A single 7B VLM fine-tuning run requires 40–80GB GPU RAM.

Engineering mindset: Each challenge has an established mitigation. Data heterogeneity → up-sampling + loss weighting. Representation alignment → projector/QFormer training. Stability → gradient clipping + warm-up. Compute → Flash Attention 2 + visual token compression.

SECTION 2 — THE THREE-STAGE PIPELINE

The Multi-Stage Training Pipeline

How all large multimodal models (GPT-4o, Gemini, LLaVA) are actually built — stage by stage

Stage 1: Unimodal Pre-training

Each encoder is pre-trained independently on large unimodal datasets — before any multimodal training begins

Image Encoder (ViT):

Pre-trained on ImageNet-21k (14M images, supervised) or LAION-2B (image-only, masked image modelling). Goal: learn to extract rich visual features — textures, objects, spatial relationships. Popular choices: CLIP ViT-L/14, EVA-ViT-G, SigLIP-400M.

Audio Encoder:

wav2vec 2.0 or HuBERT pre-trained on LibriSpeech (960 hrs, audio-only, masked prediction). Goal: learn rich speech representations without any text labels. Output: high-quality acoustic features that capture phonemes, prosody, and speaker identity.

Language Model (LLM):

GPT-4, LLaMA 3, Mistral, Phi-3 — pre-trained on text corpora (trillions of tokens, next-token prediction). Goal: learn world knowledge, reasoning, and instruction following. Will serve as the shared fusion backbone in Stage 2.

Why start unimodal? Strong per-modality representations reduce the search space for Stage 2 alignment. Re-using pretrained encoders dramatically cuts compute — training a ViT from scratch costs 10,000× more GPU-hours than using a pretrained CLIP ViT.

Stage 2: Cross-Modal Alignment Pre-training

Connect encoders via a fusion bridge (projector, QFormer, cross-attention) trained on paired multimodal data

What is trained: The projector/bridge (MLP, Q-Former, cross-attention layers). The large encoders are typically FROZEN. Training only the bridge dramatically reduces compute vs joint training.

CLIP-style Contrastive (ITC)

Match image-text pairs from LAION-400M, CC12M, WIT. Push matching pairs together in embedding space via InfoNCE. The backbone of CLIP, SigLIP, AudioCLIP. Zero-shot retrieval capability after training.

Image-Text Matching (ITM)

Binary classification: does this image match this text? Models learn to detect fine-grained alignment mismatches. Used in BLIP, ALBEF. Complements contrastive learning — catches hard negatives that ITC misses.

Generative Alignment (ITG)

Auto-regressively generate text descriptions from images (captioning objective). Forces the bridge to extract text-relevant visual features. Used in OFA, BLIP, Flamingo. The model learns to 'explain' what it sees.

Masked Cross-Modal Prediction

Mask tokens in one modality; predict from the other. BEiT-3, data2vec 2.0. Extremely data-efficient — only 14M image-text pairs needed. Hybrid with ITG for best results.

Cannot skip Stage 2: Without cross-modal alignment, the LLM backbone cannot follow visual instructions — the image encoder's output is meaningless tokens to the LLM. Stage 2 is the translation layer. Skipping it and going straight to instruction tuning produces a model that ignores images entirely.

Stage 3: Instruction Tuning for Task Alignment

After alignment, the model can process modalities but doesn't know how to help — instruction tuning teaches it

Goal: Transform an aligned multimodal model into a helpful assistant that follows natural language instructions about visual content. Fine-tune on (image, instruction, response) triplets.

LLaVA-Instruct-158K:

GPT-4 generated instruction-following data from COCO images: conversations, detailed descriptions, complex reasoning tasks. The original LLaVA training set.

ShareGPT4V-1.2M:

1.2M high-quality GPT-4V generated image-text pairs. Higher quality captions than LLaVA-Instruct. Used in LLaVA-1.5, InternVL.

ScienceQA + TextVQA:

Multimodal reasoning with scientific figures, diagrams, charts, and text in images. Teaches the model to understand structured visual content beyond natural photos.

VideoChat2-IT:

2M instruction pairs covering video question answering and temporal reasoning. Extends the recipe to video understanding — same architecture, longer sequences.

Synthetic data dominance: In 2024–2025, GPT-4o-generated synthetic instruction data has become the dominant fine-tuning strategy. Models fine-tuned on GPT-4o-generated data routinely outperform those trained on human-annotated data at lower cost.

SECTION 3 — DATA CURATION

Data Quality Is as Important as Architecture

The DataComp insight: 1.28B curated beats 5B uncured for CLIP training

Multimodal Training Datasets: Sources and Scale

Knowing your data sources is essential for understanding model capabilities and failure modes

Dataset	Scale	Content	Used For
LAION-5B	5.85B image-text pairs	Web-scraped from CommonCrawl	CLIP, Stable Diffusion pre-training
CC12M (Conceptual Captions)	12M pairs	Higher quality web alt-text	Better CLIP alignment quality
WIT (Wikipedia Image-Text)	37.5M pairs	Encyclopaedic, multilingual	Clean aligned training data
DataComp-1B	1.28B curated subset	CLIP-score filtered CC	CLIP training — quality beats quantity
WebVid-10M	10M video-caption pairs	Web video with captions	Video-language models
AudioSet	2M audio clips	527 sound categories	Audio classification, AudioCLIP

DataComp insight: DataComp-1B (1.28B CLIP-score filtered subset of CommonCrawl) outperforms the full 5B LAION-5B dataset for CLIP training. Quality beats quantity. CLIP-score filtering (discard pairs below cosine similarity threshold 0.3) is the single most impactful data quality step.

Curriculum Learning and Modality Balancing

Train in order of difficulty; balance data across modalities — two techniques that dramatically improve convergence

Curriculum Learning

Inspired by human learning: present training data in increasing difficulty.

Phase 1: Simple image-text pairs with clear visual-linguistic correspondence (single objects, short captions, high CLIP score).

Phase 2: Complex multi-object scenes with detailed descriptions.

Phase 3: Instruction-following with multi-turn conversations and reasoning.

Phase 4: Hard negatives, adversarial examples for robustness.

Benefits: Faster convergence (early epochs not wasted on impossible examples). Better generalisation. Used in LLaVA-1.6, InternVL, Cambrian-1.

Modality Balancing

Without balancing, text dominates — the model ignores visual/audio inputs:

Up-sampling: Over-sample rare modalities to equalise batch composition. 1 video-text pair = weight of 10 image-text pairs.

Loss weighting:

$$\mathcal{L} = \lambda_{\text{text}} \cdot \mathcal{L}_t + \lambda_{\text{img}} \cdot \mathcal{L}_v + \lambda_{\text{aud}} \cdot \mathcal{L}_a$$

Gradient magnitude equalisation: Normalise gradient norms from each modality head before accumulation.

Modality dropout: Randomly drop one modality with probability p during training — forces the model to work without it. Improves robustness to missing modalities at inference.

Practical impact: Modality balancing typically adds 2–4 percentage points on vision benchmarks at zero cost in architecture. It is the most underrated training technique in multimodal AI — often not mentioned in papers but critical in practice.

SECTION 4 — THE LLaVA-1.5 RECIPE

LLaVA-1.5: A \$300 Multimodal Training Recipe

The most widely replicated open-source multimodal training recipe — step-by-step with exact parameters

LLaVA-1.5 Two-Stage Training: Exact Parameters

Liu et al. (2023) — demonstrates that high-quality multimodal models can be trained for under \$300 on cloud GPUs

Stage 1 — Feature Alignment (1 epoch, ~3.5 hours on 8×A100)

Data: 558K LAION-CC-SBU image-caption pairs **What trains:** 2-layer MLP projector ONLY (LLM and ViT fully frozen) **Batch:** 256 **LR:** 1×10^{-3} **Goal:** Align image features to LLM token space

Stage 2 — Visual Instruction Tuning (1 epoch, ~20 hours on 8×A100)

Data: 665K mixed: LLaVA-Instruct + ShareGPT + ScienceQA + TextVQA **What trains:** Projector + LLM (ViT still FROZEN) **Batch:** 128 **LR:** 2×10^{-5}
Goal: Teach model to follow visual instructions

Total cost:	~\$300 on cloud A100s	Stage 1 time:	~3.5 hours	Stage 2 time:	~20 hours
Stage 1 hardware:	8×A100 80GB	Stage 2 hardware:	8×A100 80GB	Benchmarks:	VQAv2: 80.0, GQA: 63.3, MMBench: 76.3

Key Takeaways

The six essential points from this concept

Three-stage pipeline is universal Unimodal pre-training → cross-modal alignment → instruction tuning. All large multimodal models (GPT-4o, Gemini, LLaVA) follow this structure. Cannot skip Stage 2.

Three complementary objectives Contrastive (ITC) for retrieval, Image-Text Matching (ITM) for fine-grained alignment, Generative (ITG) for captioning. BLIP-2 uses all three simultaneously with a weighted sum loss.

Data quality beats quantity DataComp insight: 1.28B curated (CLIP-score filtered) outperforms 5B uncured. CLIP-score filtering, deduplication, and caption quality checks are mandatory pre-processing steps.

Modality imbalance is the primary instability Text data: trillions of tokens. Image-text: billions. Audio-visual: tens of millions. Address with up-sampling, loss weighting (λ_{img} , λ_{aud}), and modality dropout.

Curriculum learning accelerates convergence Simple pairs first, complex reasoning last. Used in LLaVA-1.6, InternVL, Cambrian-1, Gemini training recipes.

LLaVA-1.5 recipe: \$300 total Stage 1 (3.5 hrs, 8xA100): projector only, 558K captions. Stage 2 (20 hrs, 8xA100): projector+LLM, 665K instructions. Widely accessible for academic and startup use.

Note: Next: AI-MM-IN-TH-000013 — Multimodal Fine-Tuning (LoRA, QLoRA, DPO, catastrophic forgetting)

AI-MM-IN-TH-000012 Complete

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